

Xue Bin (Jason) Peng

Assistant Professor, Simon Fraser University · Research Scientist, NVIDIA

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EDUCATION

PhD in Computer Science , University of California, Berkeley Advisor: Sergey Levine and Pieter Abbeel	2017 – 2021
MSc in Computer Science , University of British Columbia Advisor: Michiel van de Panne Governor-General's Gold Medal top of master's class across all faculties (~2000 students)	2015 – 2017
Computer Science Honours , University of British Columbia Governor-General's Silver Medal in Science top of undergrad class in faculty of science (~2000 students)	2010 – 2015
School of Interactive Arts and Technology , Simon Fraser University	2009 – 2010

EMPLOYMENT

Assistant Professor , Simon Fraser University	2022 – Present
Research Scientist , NVIDIA	2022 – Present
Research Scientist Intern , NVIDIA	May 2021 – June 2022
Research Intern , Google Brain	June 2019 – May 2020
Member of Technical Staff (Intern) , OpenAI	May – Aug. 2017
Research Intern , Adobe Research	May – Aug. 2015
Lab Associate (Intern) , Disney Research Pittsburgh	Jan. – May 2015
Intern Software Developer , Microsoft Studios	May – Nov. 2013
Co-op Rendering Engineer , Capcom Vancouver	Jan. – Aug. 2012

AWARDS

Early Career Researcher Award , CHCCS/SCDHM Graphics Interface	2024
Outstanding Doctoral Dissertation Award , ACM SIGGRAPH	2022
Sevin Rosen Funds Award for Innovation , University of California, Berkeley	2021
Berkeley Fellowship For Graduate Study , University of California, Berkeley	2017 – 2020
NSERC Postgraduate Scholarship , University of California, Berkeley	2017 – 2020
Governor-General's Gold Medal in Science , University of British Columbia top of master's class across all faculties (~2000 students)	2017
NSERC Canada Graduate Scholarship Master's Award , University of British Columbia	2017
Theodore E Arnold Fellowship , University of British Columbia	2015 – 2016

CS Merit Award , University of British Columbia	2015 – 2017
Governor-General's Silver Medal in Science , University of British Columbia top of undergraduate class in faculty of science (~2000 students)	2015
Greer Family Scholarship , University of British Columbia	2013
Marie Kendall Memorial Scholarship in Science , University of British Columbia	2013
Charles and Jane Banks Scholarship , University of British Columbia	2011
Computer Science Scholarship , University of British Columbia	2011
Trek Excellence Scholarship , University of British Columbia	2011 – 2015
Norman A M MacKenzie Scholarship , University of British Columbia	2010
President's Entrance Scholarship , University of British Columbia	2010
Gordon M. Shrum Scholarship , Simon Fraser University	2009 – 2010

PUBLICATIONS

Refereed Journals/Conferences

2026

- [1] Sunmin Lee, Davis Rempe, Yifeng Jiang, Haotian Zhang, Tingwu Wang, Jungdam Won, and **Xue Bin Peng**. "DMP: Directable Motion Retargeting through Motion Paraphrasing". In: *ACM Trans. Graph.* 46.1 (2026).
- [2] Jiashun Wang, Yifeng Jiang, Haotian Zhang, Chen Tessler, Davis Rempe, Jessica Hodgins, and **Xue Bin Peng**. "HIL: Hybrid Imitation Learning for Dynamic Athletic Control". In: *ACM Trans. Graph.* (2026).
- [3] Yuxuan Mu, Ziyu Zhang, Yi Shi, Dun Yang, Minami Matsumoto, Kotaro Imamura, Guy Tevet, Chuan Guo, Michael Taylor, Chang Shu, Pengcheng Xi, and **Xue Bin Peng**. "SMP: Reusable Score-Matching Motion Priors for Physics-Based Character Control". In: *ACM Transactions on Graphics (Proceedings of SIGGRAPH 2026)* (2026).
- [4] Yi Shi, Yifeng Jiang, Chen Tessler, and **Xue Bin Peng**. "GPC: Large-Scale Generative Pretraining for Transferable Motor Control". In: *SIGGRAPH '26 Conference Proceedings*. 2026.
- [5] Tingwu Wang, Olivier Dionne, Michael de Ruyter, David Minor, Davis Rempe, Kaifeng Zhao, Mathis Petrovich, Ye Yuan, Chenran Li, Zhengyi Luo, Brian Robison, Xavier Blackwell, Bernardo Antoniazzi, **Xue Bin Peng**, Yuke Zhu, and Simon Yuen. "MotionBricks: Scalable Real-Time Motions with Modular Latent Generative Model and Smart Primitives". In: *ACM Trans. Graph.* 45.4 (2026).
- [6] Nam Hee Kim, Jingjing May Liu, Jaakko Lehtinen, Perttu Hämäläinen, James F. O'Brien, and **Xue Bin Peng**. "Robo-Saber: Generating and Simulating Virtual Reality Players". In: *Computer Graphics Forum* 45.2 (2026), e70353.
- [7] Zhaoyuan Gu, Junheng Li, Wenlan Shen, Wenhao Yu, Zhaoming Xie, Stephen McCrory, Xianyi Cheng, Abdulaziz Shamsah, Robert Griffin, C. Karen Liu, Abderrahmane Kheddar, **Xue Bin Peng**, Yuke Zhu, Guanya Shi, Quan Nguyen, Gordon Cheng, Huijun Gao, and Ye Zhao. "Humanoid Locomotion and Manipulation: Current Progress and Challenges in Control, Planning, and Learning". In: *IEEE/ASME Transactions on Mechatronics* 31.2 (2026), pp. 2300–2330.

2025

- [8] Chen Tessler, Yifeng Jiang, Erwin Coumans, Zhengyi Luo, Gal Chechik, and **Xue Bin Peng**. “MaskedManipulator: Versatile Whole-Body Manipulation”. In: *ACM SIGGRAPH Asia 2025 Conference Proceedings*. 2025.
- [9] Ziyu Zhang, Sergey Bashkirov, Dun Yang, Yi Shi, Michael Taylor, and **Xue Bin Peng**. “Physics-Based Motion Imitation with Adversarial Differential Discriminators”. In: *SIGGRAPH Asia 2025 Conference Papers (SIGGRAPH Asia '25 Conference Papers)*. 2025.
- [10] Yuxuan Mu, Hung Yu Ling, Yi Shi, Ismael Baira Ojeda, Pengcheng Xi, Chang Shu, Fabio Zinno, and **Xue Bin Peng**. “StableMotion: Training Motion Cleanup Models with Unpaired Corrupted Data”. In: *SIGGRAPH Asia 2025 Conference Papers (SIGGRAPH Asia '25 Conference Papers)*. 2025.
- [11] Yanjie Ze, Zixuan Chen, Joao Pedro Araujo, Zi-ang Cao, **Xue Bin Peng**, Jiajun Wu, and Karen Liu. “TWIST: Teleoperated Whole-Body Imitation System”. In: *Proceedings of The 9th Conference on Robot Learning*. Vol. 305. 2025, pp. 2143–2154.
- [12] Zixuan Chen, Xialin He, Yen-Jen Wang, Qiayuan Liao, Yanjie Ze, Zhongyu Li, S. Shankar Sastry, Jiajun Wu, Koushil Sreenath, Saurabh Gupta, and **Xue Bin Peng**. “Learning Smooth Humanoid Locomotion through Lipschitz-Constrained Policies”. In: *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2025.
- [13] Yanjie Ze, Zixuan Chen, Wenhao Wang, Tianyi Chen, Xialin He, Ying Yuan, **Xue Bin Peng**, and Jiajun Wu. “Generalizable Humanoid Manipulation with 3D Diffusion Policies”. In: *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2025, pp. 2873–2880.
- [14] Michael Xu, Yi Shi, KangKang Yin, and **Xue Bin Peng**. “PARC: Physics-based Augmentation with Reinforcement Learning for Character Controllers”. In: *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers*. 2025.
- [15] Guy Tevet, Sigal Raab, Setareh Cohan, Daniele Reda, Zhengyi Luo, **Xue Bin Peng**, Amit Haim Bermano, and Michiel van de Panne. “CLoSD: Closing the Loop between Simulation and Diffusion for multi-task character control”. In: *The Thirteenth International Conference on Learning Representations*. 2025. **Spotlight**.
- [16] Xiaoyu Huang, Yufeng Chi, Ruofeng Wang, Zhongyu Li, **Xue Bin Peng**, Sophia Shao, Borivoje Nikolic, and Koushil Sreenath. “DiffuseLoco: Real-Time Legged Locomotion Control with Diffusion from Offline Datasets”. In: *Proceedings of The 8th Conference on Robot Learning*. Vol. 270. 2025, pp. 1567–1589.

2024

- [17] Zhongyu Li, **Xue Bin Peng**, Pieter Abbeel, Sergey Levine, Glen Berseth, and Koushil Sreenath. “Reinforcement learning for versatile, dynamic, and robust bipedal locomotion control”. In: *The International Journal of Robotics Research* (2024), p. 02783649241285161.
- [18] Xiaoyu Huang, Qiayuan Liao, Yiming Ni, Zhongyu Li, Laura Smith, Sergey Levine, **Xue Bin Peng**, and Koushil Sreenath. “HiLMa-Res: A General Hierarchical Framework via Residual RL for Combining Quadrupedal Locomotion and Manipulation”. In: *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2024, pp. 9050–9057.

- [19] Chen Tessler, Yunrong Guo, Ofir Nabati, Gal Chechik, and **Xue Bin Peng**. “MaskedMimic: Unified Physics-Based Character Control Through Masked Motion Inpainting”. In: *ACM Transactions on Graphics (TOG)* (2024).
- [20] Yi Shi, Jingbo Wang, Xuekun Jiang, Bingkun Lin, Bo Dai, and **Xue Bin Peng**. “Interactive Character Control with Auto-Regressive Motion Diffusion Models”. In: *ACM Trans. Graph.* 43 (2024).
- [21] Jordan Juravsky, Yunrong Guo, Sanja Fidler, and **Xue Bin Peng**. “SuperPADL: Scaling Language-Directed Physics-Based Control with Progressive Supervised Distillation”. In: *SIGGRAPH 2024 Conference Papers (SIGGRAPH '24 Conference Papers)*. 2024.
- [22] Setareh Cohan, Guy Tevet, Daniele Reda, **Xue Bin Peng**, and Michiel van de Panne. “Flexible Motion In-betweening with Diffusion Models”. In: *ACM SIGGRAPH 2024 Conference Proceedings*. 2024.
- [23] Hongwei Yi, Justus Thies, Michael J. Black, **Xue Bin Peng**, and Davis Rempe. “Generating Human Interaction Motions in Scenes with Text Control”. In: *European Conference on Computer Vision (ECCV)*. 2024.
- [24] Mathis Petrovich, Or Litany, Umar Iqbal, Michael J. Black, Gül Varol, **Xue Bin Peng**, and Davis Rempe. “Multi-Track Timeline Control for Text-Driven 3D Human Motion Generation”. In: *CVPR Workshop on Human Motion Generation*. 2024.
- [25] Jonah Philion, **Xue Bin Peng**, and Sanja Fidler. “Trajenglish: Traffic Modeling as Next-Token Prediction”. In: *The Twelfth International Conference on Learning Representations*. 2024.

2023

- [26] Haotian Zhang, Ye Yuan, Viktor Makoviyshuk, Yunrong Guo, Sanja Fidler, **Xue Bin Peng**, and Kayvon Fatahalian. “Learning Physically Simulated Tennis Skills from Broadcast Videos”. In: *ACM Trans. Graph.* 42.4 (2023). **Best Paper Honourable Mention**.
- [27] Mohamed Hassan, Yunrong Guo, Tingwu Wang, Michael J. Black, Sanja Fidler, and **Xue Bin Peng**. “Synthesizing Physical Character-Scene Interactions”. In: *ACM SIGGRAPH 2023 Conference Proceedings*. 2023.
- [28] Chen Tessler, Yoni Kasten, Yunrong Guo, Shie Mannor, Gal Chechik, and **Xue Bin Peng**. “CALM: Conditional Adversarial Latent Models for Directable Virtual Characters”. In: *ACM SIGGRAPH 2023 Conference Proceedings*. 2023.
- [29] Alejandro Escontrela, Ademi Adeniji, Wilson Yan, Ajay Jain, **Xue Bin Peng**, Ken Goldberg, Youngwoon Lee, Danijar Hafner, and Pieter Abbeel. “Video Prediction Models as Rewards for Reinforcement Learning”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 36. 2023.
- [30] Xiaoyu Huang, Zhongyu Li, Yanzhen Xiang, Yiming Ni, Yufeng Chi, Yunhao Li, Lizhi Yang, **Xue Bin Peng**, and Koushil Sreenath. “Creating a dynamic quadrupedal robotic goalkeeper with reinforcement learning”. In: *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2023, pp. 2715–2722.
- [31] Laura M. Smith, J. Chase Kew, Tianyu Li, Linda Luu, **Xue Bin Peng**, Sehoon Ha, Jie Tan, and Sergey Levine. “Learning and Adapting Agile Locomotion Skills by Transferring Experience”. In: *Robotics: Science and Systems XIX, Daegu, Republic of Korea, July 10-14, 2023*. 2023.
- [32] Kevin Zakka, Philipp Wu, Laura Smith, Nimrod Gileadi, Taylor Howell, **Xue Bin Peng**, Sumeet Singh, Yuval Tassa, Pete Florence, Andy Zeng, and Pieter Abbeel. “RoboPianist: Dexterous Piano Playing with Deep Reinforcement Learning”. In: *Proceedings of The 7th Conference on Robot Learning*. Vol. 229. 2023, pp. 2975–2994.

- [33] Davis Rempe, Zhengyi Luo, **Xue Bin Peng**, Ye Yuan, Kris Kitani, Karsten Kreis, Sanja Fidler, and Or Litany. "Trace and Pace: Controllable Pedestrian Animation via Guided Trajectory Diffusion". In: *Conference on Computer Vision and Pattern Recognition (CVPR)*. 2023.
- [34] Zhongyu Li, **Xue Bin Peng**, Pieter Abbeel, Sergey Levine, Glen Berseth, and Koushil Sreenath. "Robust and Versatile Bipedal Jumping Control through Reinforcement Learning". In: *Robotics: Science and Systems XIX, Daegu, Republic of Korea, July 10-14, 2023*. 2023.
- [35] Gilbert Feng, Hongbo Zhang, Zhongyu Li, **Xue Bin Peng**, Bhuvan Basireddy, Linzhu Yue, ZHITAO SONG, Lizhi Yang, Yunhui Liu, Koushil Sreenath, and Sergey Levine. "GenLoco: Generalized Locomotion Controllers for Quadrupedal Robots". In: *Proceedings of The 6th Conference on Robot Learning*. Vol. 205. 2023, pp. 1893–1903.

2022

- [36] Michael Laskin, Hao Liu, **Xue Bin Peng**, Denis Yarats, Aravind Rajeswaran, and Pieter Abbeel. "Unsupervised Reinforcement Learning with Contrastive Intrinsic Control". In: *Advances in Neural Information Processing Systems*. 2022.
- [37] Jordan Juravsky, Yunrong Guo, Sanja Fidler, and **Xue Bin Peng**. "PADL: Language-Directed Physics-Based Character Control". In: *SIGGRAPH Asia 2022 Conference Papers*. 2022.
- [38] Alejandro Escontrela, **Xue Bin Peng**, Wenhao Yu, Tingnan Zhang, Atil Iscen, Ken Goldberg, and Pieter Abbeel. "Adversarial Motion Priors Make Good Substitutes for Complex Reward Functions". In: *International Conference on Intelligent Robots and Systems (IROS)*. 2022.
- [39] Yandong Ji, Zhongyu Li, Yinan Sun, **Xue Bin Peng**, Sergey Levine, Glen Berseth, and Koushil Sreenath. "Hierarchical Reinforcement Learning for Precise Soccer Shooting Skills using a Quadrupedal Robot". In: *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2022, pp. 1479–1486.
- [40] **Xue Bin Peng**, Yunrong Guo, Lina Halper, Sergey Levine, and Sanja Fidler. "ASE: Large-scale Reusable Adversarial Skill Embeddings for Physically Simulated Characters". In: *ACM Trans. Graph.* 41.4 (2022).
- [41] Laura Smith, J. Chase Kew, **Xue Bin Peng**, Sehoon Ha, Jie Tan, and Sergey Levine. "Legged Robots that Keep on Learning: Fine-Tuning Locomotion Policies in the Real World". In: *2022 IEEE International Conference on Robotics and Automation (ICRA)*. 2022, pp. 1593–1599.

2021

- [42] Seungmoon Song, Łukasz Kidziński, **Xue Bin Peng**, Carmichael Ong, Jennifer Hicks, Sergey Levine, Christopher G. Atkeson, and Scott L. Delp. "Deep reinforcement learning for modeling human locomotion control in neuromechanical simulation". In: *Journal of NeuroEngineering and Rehabilitation* 18.1 (2021), p. 126.
- [43] Eric Mitchell, Rafael Rafailov, **Xue Bin Peng**, Sergey Levine, and Chelsea Finn. "Offline Meta-Reinforcement Learning with Advantage Weighting". In: *Proceedings of the 38th International Conference on Machine Learning*. 2021.
- [44] **Xue Bin Peng**, Ze Ma, Pieter Abbeel, Sergey Levine, and Angjoo Kanazawa. "AMP: Adversarial Motion Priors for Stylized Physics-Based Character Control". In: *ACM Trans. Graph.* 40.4 (2021).

- [45] Zhongyu Li, Xuxin Cheng, **Xue Bin Peng**, Pieter Abbeel, Sergey Levine, Glen Berseth, and Koushil Sreenath. “Reinforcement Learning for Robust Parameterized Locomotion Control of Bipedal Robots”. In: *2021 IEEE International Conference on Robotics and Automation (ICRA)*. 2021, pp. 2811–2817.

2020

- [46] **Xue Bin Peng**, Erwin Coumans, Tingnan Zhang, Tsang-Wei Edward Lee, Jie Tan, and Sergey Levine. “Learning Agile Robotic Locomotion Skills by Imitating Animals”. In: *Robotics: Science and Systems*. 2020. **Best Paper**.
- [47] Anirudh Goyal, Shagun Sodhani, Jonathan Binas, **Xue Bin Peng**, Sergey Levine, and Yoshua Bengio. “Reinforcement Learning with Competitive Ensembles of Information-Constrained Primitives”. In: *International Conference on Learning Representations*. 2020.

2019

- [48] Farzad Abdolhosseini, Hung Yu Ling, Zhaoming Xie, **Xue Bin Peng**, and Michiel van de Panne. “On Learning Symmetric Locomotion”. In: *Motion, Interaction and Games*. 2019.
- [49] **Xue Bin Peng**, Michael Chang, Grace Zhang, Pieter Abbeel, and Sergey Levine. “MCP: Learning Composable Hierarchical Control with Multiplicative Compositional Policies”. In: *Advances in Neural Information Processing Systems* 32. 2019, pp. 3681–3692.
- [50] **Xue Bin Peng**, Angjoo Kanazawa, Sam Toyer, Pieter Abbeel, and Sergey Levine. “Variational Discriminator Bottleneck: Improving Imitation Learning, Inverse RL, and GANs by Constraining Information Flow”. In: *International Conference on Learning Representations*. 2019.

2018

- [51] **Xue Bin Peng**, Angjoo Kanazawa, Jitendra Malik, Pieter Abbeel, and Sergey Levine. “SFV: Reinforcement Learning of Physical Skills from Videos”. In: *ACM Trans. Graph.* 37.6 (2018).
- [52] **Xue Bin Peng**, Pieter Abbeel, Sergey Levine, and Michiel van de Panne. “DeepMimic: Example-guided Deep Reinforcement Learning of Physics-based Character Skills”. In: *ACM Trans. Graph.* 37.4 (2018), 143:1–143:14.
- [53] **Xue Bin Peng**, Marcin Andrychowicz, Wojciech Zaremba, and Pieter Abbeel. “Sim-to-Real Transfer of Robotic Control with Dynamics Randomization”. In: *2018 IEEE International Conference on Robotics and Automation (ICRA)*. 2018, pp. 3803–3810.

2017

- [54] **Xue Bin Peng**, Glen Berseth, Kangkang Yin, and Michiel Van De Panne. “DeepLoco: Dynamic Locomotion Skills Using Hierarchical Deep Reinforcement Learning”. In: *ACM Trans. Graph.* 36.4 (2017), 41:1–41:13.
- [55] **Xue Bin Peng** and Michiel van de Panne. “Learning Locomotion Skills Using DeepRL: Does the Choice of Action Space Matter?” In: *Proceedings of the ACM SIGGRAPH / Eurographics Symposium on Computer Animation*. 2017, 12:1–12:13. **Best Student Paper**.

2016

- [56] **Xue Bin Peng**, Glen Berseth, and Michiel van de Panne. “Terrain-adaptive Locomotion Skills Using Deep Reinforcement Learning”. In: *ACM Trans. Graph.* 35.4 (2016), 81:1–81:12.

2015

- [57] **Xue Bin Peng**, Glen Berseth, and Michiel van de Panne. “Dynamic Terrain Traversal Skills Using Reinforcement Learning”. In: *ACM Trans. Graph.* 34.4 (2015), 80:1–80:11.

Non-Refereed

- [58] Zewei Zhang, Kehan Wen, Michael Xu, Junzhe He, Chenhao Li, Takahiro Miki, Clemens Schwarke, Chong Zhang, **Xue Bin Peng**, and Marco Hutter. *Learning Whole-Body Humanoid Locomotion via Motion Generation and Motion Tracking*. 2026. arXiv: [2604.17335 \[cs.R0\]](#).
- [59] Davis Rempe, Mathis Petrovich, Ye Yuan, Haotian Zhang, **Xue Bin Peng**, Yifeng Jiang, Tingwu Wang, Umar Iqbal, David Minor, Michael de Ruyter, Jiefeng Li, Chen Tessler, Edy Lim, Eugene Jeong, Sam Wu, Ehsan Hassani, Michael Huang, Jin-Bey Yu, Chaeyeon Chung, Lina Song, Olivier Dionne, Jan Kautz, Simon Yuen, and Sanja Fidler. *Kimodo: Scaling Controllable Human Motion Generation*. 2026. arXiv: [2603.15546 \[cs.CV\]](#).
- [60] Aviral Kumar, **Xue Bin Peng**, and Sergey Levine. “Reward-Conditioned Policies”. In: *arXiv preprint arXiv:1912.13465* (2019).
- [61] **Xue Bin Peng**, Aviral Kumar, Grace Zhang, and Sergey Levine. “Advantage-Weighted Regression: Simple and Scalable Off-Policy Reinforcement Learning”. In: *CoRR abs/1910.00177* (2019). arXiv: [1910.00177](#).

PATENTS

Generating motion from text in content generation systems and applications

Xue Bin Peng, Jonathan Tseng, Davis Winston Rempe, Or Litany, Ye Yuan, Umar Iqbal, Sanja Fidler, Jan Kautz
US Patent Application: 18819169 (2026)

Global Human and Camera Motion Estimation with Motion Diffusion Model

Jiefeng Li, Umar Iqbal, Ye Yuan, Davis Rempe, Jan Kautz, Haotian Zhang, Xue Bin Peng, Pavlo Molchanov
US Patent Application: 18795561 (2025)

Scene-Aware Synthetic Human Motion Generation Using Neural Networks

Yi Hongwei, Davis Rempe, Xue Bin Peng, Sanja Fidler
US Patent Application: 18415496 (2025)

Generative Human Motion Simulation with Temporal Control

Mathis Petrovich, Xue Bin Peng, Davis Rempe, Umar Iqbal, Or Litany, Sanja Fidler
US Patent Application: 18404758 (2025)

Virtual Agent Trajectory Prediction and Traffic Modeling for Machine Simulation Systems and Applications

Jonah Phillion, Jeevan Devaranjan, Xue Bin Peng, Sanja Fidler
US Patent Application: 17949991 (2024)

Object Animation Using Neural Networks

Jordan Juravsky, Xue Bin Peng, Sanja Fidler
US Patent Application: 17748739 (2024)

Physics-Based Image Generation Using One or More Neural Networks

Tingwu Wang, Yunrong Guo, Xie Cheng, Xue Bin Peng, Sanja Fidler

US Patent Application: 17842481 (2023)

INVITED TALKS

Motion Imitation for Versatile and Lifelike Motor Skills

July 2, 2026

VinMotion

Hanoi, Vietnam

Synthetic Motion Data for Versatile Humanoid Control

June 1, 2026

ICRA 2026 Workshop: Synthetic Data for Robot Learning

Vienna, Austria

Beyond Motion Tracking for Motor Skill Learning

May 14, 2026

Wayve

Vancouver, Canada

Learning Whole-Body Motion Control With Adversarial Differential Discriminators

April 19, 2026

China3DV

Hangzhou, China

Beyond Motion Tracking For Versatile and Lifelike Motor Skills

April 17, 2026

Unitree Robotics

Hangzhou, China

Scaling Up Motion Imitation for Versatile Whole-Body Control

December 18, 2025

TriFusion: Towards Embodied Intelligence Across Humans, Avatars, and Humanoid Robotics

SIGGRAPH Asia 2025 Technical Workshop

Hong Kong, China

Beyond Motion Tracking for Motor Skill Learning

August 11, 2025

SIGGRAPH Technical Workshop: Generalizing Natural Behavior: Retargeting Human or Animal Motion to Robotic Forms.

Vancouver, British Columbia, Canada

Beyond Motion Tracking for Motor Skill Learning

July 31, 2025

Amazon

Emeryville, California, USA

Beyond Motion Tracking for Motor Skill Learning

June 11, 2025

Workshop on Humanoid Agents at CVPR

Nashville, Tennessee, USA

Beyond Motion Tracking for Motor Skill Learning

June 2, 2025

Stanford University

Stanford, California, USA

Acquiring Motor Skills Through Motion Imitation and Reinforcement Learning

June 3, 2024

Graphics Interfaces 2024

Halifax, Canada

Acquiring Motor Skills Through Motion Imitation and Reinforcement Learning

May 3, 2024

University of Alberta

Alberta, Canada

Acquiring Motor Skills Through Motion Imitation and Reinforcement Learning

October 19, 2023

Fourier Intelligence Co., Ltd

Shanghai, China

Learning to Move from Videos

August 23, 2023

Electronic Arts Inc.

Burnaby, British Columbia, Canada

Adversarial Imitation Learning for Motor Control

August 7, 2022

SFU Visual Computing Workshop

Vancouver, British Columbia, Canada

Imitation Learning for Data-Driven Physics-Based Character Control

April 22, 2021

Adobe Inc.

San Jose, California, USA

Data-Driven Physics-Based Character Animation with Imitation Learning

March 24, 2021

Peking University

Beijing, China

Learning Agile Robotic Locomotion Skills by Imitating Animals

November 2, 2020

New York University

New York, New York, USA

Data-Driven Physics-Based Character Animation with Imitation Learning

February 21, 2020

Electronic Arts Inc.

Redwood City, California, USA

Data-Driven Physics-Based Character Animation with Imitation Learning

July 23, 2019

Bellairs Workshop

Folkestone, St. James, Barbados

Towards a Virtual Stuntman

July 23, 2018

Boston Dynamics

Waltham, Massachusetts, USA

Developing Locomotion Skills with Reinforcement Learning

March 14, 2017

Google LLC

Mountain View, California, USA

ADVISING

Current

Yi Shi, Ph.D. student

2023 – Present

Dun Yang, Ph.D. student	2024 – Present
Yuxuan Mu, Ph.D. student	2024 – Present
Ziyu (Charlotte) Zhang, Ph.D. student	2024 – Present
Sunwoo Kim, Ph.D. student	2025 – Present
Ruiyu Gou, Ph.D. student	2025 – Present
Nicholas Ioannidis, Ph.D. student	2026 – Present

Alumni

Michael Xu, M.Sc., AI Resident at 1X	2023 – 2025
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TEACHING

Instructor , Simon Fraser University	2022 – Present
CMPT 361: Intro to Visual Computing	
CMPT 729: Reinforcement Learning	